Adam Boesky

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Education

***Candidate* PhD, Harvard University** *Cambridge, MA* *Aug 2026 – Present*
Subject: Astrophysics

BA, Harvard University *Cambridge, MA* *Aug 2021 – May 2025*
Majors: Astrophysics, Mathematics *Honors:* *cum laude*
Relevant Coursework: General Relativity, Information Theory, Machine Learning for Astrophysicists, Advanced Scientific Computing, Differential Geometry, Dynamical Systems

Professional & Research Experience

ProDex Labs *Brooklyn, NY* *Mar 2025 – Present*
Founding Engineer
- Develop core general purpose discrete-event simulation engine for a digital twin models of manufacturing settings
- Implement advanced planning and scheduling algorithms for optimizing manufacturing operations
- Architect and implement tools for an AI-native agent to copilot all modules our application
- Led production and design of UX; responsible for scalable React/TypeScript frontend and Python-FastAPI backend

Center for Astrophysics | Harvard & Smithsonian *Cambridge, MA*
Berger Cosmic Transient Group *Jun 2022 – Present*
- Investigated the evolution and demographics of black holes and neutron stars with population synthesis simulations of massive binary stars
- For senior thesis: created catalog of long-duration transients with novel source extraction methods; studying the detectability of long-duration transients with LSST and *Roman*

Villar Time-Domain Astronomy Data Lab *Aug 2023 – Present*
- Developed a machine learning pipeline to rapidly classify supernovae using their inferred host galaxy's properties; preliminary website here

SpaceX *Los Angeles, CA* *Summers 2023, 2024*
Guidance, Navigation, & Control Engineering Intern
- Developed physical and empirical models for high-fidelity Monte Carlo simulation of the Falcon rocket
- Drove operational changes to promote launch vehicle reliability and performance
- Automated day of launch operations and implemented tools to accelerate launch cadence
  - Conducted study to assess the value of mission-specific analyses required for launch

Harvard School of Engineering & Applied Sciences *Cambridge, MA*
Ramanathan Lab *Jan 2023 – Jul 2024*
- Developed a biologically-realizable computational architecture for basic intelligence

Harvard C3 Policing Research Team *Jun 2020 – Aug 2022*

- Quantified impact of novel Counter Criminal Continuum (C3) policing method; presented results to Massachusetts government and law enforcement officials
- Identified statistically significant positive trends in business, crime, education, health, and housing data

Open-Source Software

astro_SPLASH — *Lead Developer* (1 stars / 1 forks)
Supernova classification Pipeline Leveraging Attributes of Supernova Hosts (SPLASH): A machine learning pipeline that classifies supernovae by first inferring their host's properties

COMPAS — *Developer* (79 stars / 75 forks)
COMPAS rapid binary population synthesis code [docs]

HOOTSim — *Lead Developer* (2 stars / 0 forks)
N-Body Simulator

Publications

refereed: 3 / first author: 3 / citations: 37 / h-index: 4 (2025-10-29)

Refereed publications

3 Team Compas; Mandel, Ilya; Riley, Jeff; **Boesky, Adam**; *et al.*, 2025, *Rapid Stellar and Binary Population Synthesis with COMPAS: Methods Paper II*, The Astrophysical Journal Supplement Series, **280**, 43 (arXiv:2506.02316) [4 citations]

2 **Boesky, Adam**; Broekgaarden, Floor S.; & Berger, Edo, 2024, *Investigating the Cosmological Rate of Compact Object Mergers from Isolated Massive Binary Stars*, The Astrophysical Journal, **976**, 24 (arXiv:2405.01630) [14 citations]

1 **Boesky, Adam**; Broekgaarden, Floor S.; & Berger, Edo, 2024, *The Binary Black Hole Merger Rate Deviates from the Cosmic Star Formation Rate: A Tug of War between Metallicity and Delay Times*, The Astrophysical Journal, **976**, 23 (arXiv:2405.01623) [13 citations]

Preprints & white papers

3 Schuetz, Ann-Kathrin; Migala, Alexander; **Boesky, Adam**; Poon, Alan W. P.; *et al.*, 2025, *RESOLVE: Rare Event Surrogate Likelihood for Gravitational Wave Paleontology Parameter Estimation*, ArXiv (arXiv:2506.00757) [1 citation]

2 **Boesky, Adam**; Villar, V. Ashley; Gagliano, Alexander; & Hsu, Brian, 2025, *SPLASH: A Rapid Host-Based Supernova Classifier for Wide-Field Time-Domain Surveys*, ArXiv (arXiv:2506.00121) [4 citations]

1 Li, Chenguang; Brenner, Jonah; **Boesky, Adam**; Ramanathan, Sharad; & Kreiman, Gabriel, 2024, *Neuron-level Prediction and Noise can Implement Flexible Reward-Seeking Behavior*, bioRxiv [1 citation]

Teaching & Mentorship

Teaching Assistant — Harvard Physics 143a: Quantum Mechanics I Spring 2024
Teaching Assistant — Harvard Physics 12b: Electromagnetism and Statistical Physics Fall 2023

I am currently mentoring and have mentored several students on a variety of projects including the following:
High schoolers: Khushi Karthikeyan (Mar 2024 — Present), Meera Desawale (Jul 2024 — Present), Daniel Garrett-Metz (Apr 2021 — Jun 2021)

Honors, Awards, and Fellowships

Honorable Mention NSF Graduate Research Funding Program, 2025
Finalist Hertz Fellowship, 2025
Finalist Rhodes Scholarship, 2025
Finalist Marshall Scholarship, 2025
Harvard Program for Research in Science and Engineering Fellow ($5000), 2022
Harvard College Research Program Fellow ($1000), 2022

Talks & Presentations

Oral
- Orange County Astronomers, Orange, CA *Upcoming* Jan 2026
- American Physical Society April Meeting, Sacramento, CA Apr 2024
- Post-Pax-VIII, Cambridge, MA Aug 2022

Poster
- American Astronomical Society Meeting, National Harbor, MD Jan 2025
- Computational and Systems Neuroscience, Lisbon, Portugal Mar 2024
- Giant Magellan Telescope 2022 Science Meeting, Sedona, AZ Sep 2022

Skills & Interests

Programming: (Expert) Python, C++, Java, Mathematica, JavaScript, SQL (Intermediate) R (Familiar) Julia
Software: Git/GitHub (>500 contributions, 20 repositories), Slurm, LaTeX, Linux, Bash, SAOImage DS9
Miscellaneous: Billiards, powerlifting, running, mixology, knitting, mountaineering, skiing, surfing, foosball